

A Cost-Effective Arduino-Based System for Measuring Solar Panel Stability

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Abstract

As the quality of perovskite solar panels improves, verifying their long-term stability becomes increasingly important. However, existing commercial testing solutions are prohibitively expensive for conducting long-term stability studies on many cells in parallel. The ability to test multiple solar cells with varying chemical compositions and structural configurations simultaneously is crucial to accelerating material optimization.

This paper presents a novel, open-source, low-cost approach to long-term stability testing using an Arduino-based system. The proposed system enables simultaneous monitoring of eight solar cells for under \$50 (INSERT PRICE) and can be easily expanded to accommodate more panels. This approach significantly reduces the cost barrier for large-scale stability studies, making extended testing feasible for research and development settings.

1 Introduction

Over the past decade, perovskite solar panels have emerged as a promising alternative to traditional silicon based photovoltaics (PV) due to their high efficiency, low production costs, and potential for environmentally friendly manufacturing. However, one of the most significant barriers to widespread commercial adoption is their limited long term stability. While traditional solar cells can operate for decades, perovskite-based cells degrade extremely rapidly due to environmental factors such as moisture, heat, and ultraviolet radiation [1]. Accurate and cost-effective stability testing is essential to improve perovskite performance for use in commercial applications.

Current commercial solutions for long term stability testing are prohibitively expensive, often costing tens of thousands of dollars per unit. These high costs make it difficult for university researchers and small companies to conduct large-scale parallel testing across multiple solar cells with different perovskite compositions and structural configurations. As a result, many labs are limited in their ability to evaluate perovskite degradation mechanisms and optimize the panels for long term stability. A low-cost, scalable, and fully open-source testing platform would significantly enhance the accessibility and ease of perovskite research.

This paper presents an Arduino-based stability measurement system that is designed to provide an affordable and easily expandable alternative to commercial stability testing equipment. The system uses an Arduino Nano Every microcontroller with a custom circuit board. The board features a MCP4725 digital-to-analog converter (DAC) for load application and an INA219 analog-to-digital converter (ADC) for voltage and current monitoring. The system is capable of performing JV curve plotting and maximum power point tracking (MPPT) for up to eight solar cells in parallel, all for under \$50 (insert price) per unit. This significantly reduces the cost barrier for large-scale perovskite stability testing, making massively parallel measurements feasible in academic and research settings.

The software component, Stability Setup Graphical User Interface (GUI), provides an intuitive interface for system control and data visualization. GUI is built entirely in python, using PySide6 library for the frontend connected to a python backend. The embedded firmware is programmed in C++ code for the Arduino. Although this paper does not focus on the development of the software suite, it will outline the software structure and user functionality to provide a complete understanding of the system's operation.

To validate the system, experiment results are compared against a commercial MPPT stability measurement system., assessing accuracy, reliability, and cost effectiveness. This comparison ensures that the proposed solution provides a viable, low-cost alternative to high-cost stability testing methods while maintaining sufficient precision for research applications.

A key feature of this testing platform is its commitment to open-source principles. The hardware schematics, pcb layout, firmware, and software are freely available, allowing researchers and developers to modify and improve upon the platform to suit their needs. This project aims to give researchers an accessible tool for advancing perovskite solar technology.

2 System Description

2.1 Hardware

2.1.1 Components:

- Arduino Nano Every (ANE) Digikey Link

- Custom Circuit Board: OSHWLab Link
- 3-D Printed Enclosure: OnShape Link
- Windows 10 or 11 Computer (PC)
- Solar Cell Holder
- Solar Simulator
- N2 supply

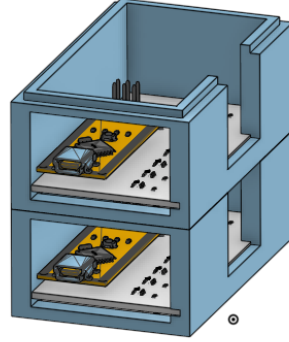


Figure 1: CAD Rendering of two systems stacked together

The system is designed to facilitate long-term stability testing of perovskite solar panels by measuring voltage and current characteristics over time. The ANE serves as the microcontroller, communicating with the custom circuit board over I²C to control measurement. The PC is used to process and visualize the collected data. The 3-D printed enclosure provides structural support and modularity for multiple stacked units.

2.1.2 Connectivity & Communication

Each ANE connects to the windows computer via a microUSB cable. This cable serves as a power source and data communication link. Depending on the PC, a powered USB expansion hub may be required to power more than 8 ANEs. The PC is used to control the ANE to start and stop measurements. A single PC is capable of handling up to 8-32 (depending on PC hardware and Windows version) stacked ANE units. This means 8-32 solar panel stability measurements running in parallel. Each board operates independently while communicating with the main PC for data logging. The PC also processes and visualizes the collected data.

2.1.3 Circuit Board

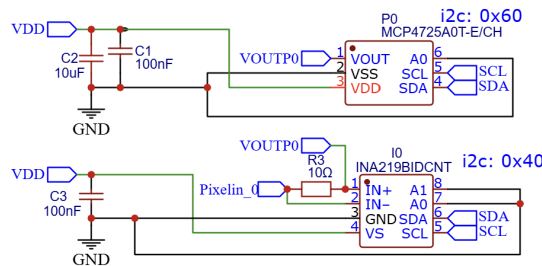


Figure 2: Schematic for measuring a single pixel

The PCB was designed in EasyEDA with parts that are easily available for assembly from JLCPCB. The schematics and pcb layout are all open sourced. It was originally designed with breakout boards from the INA219 and MCP4725 from Adafruit for ease of use and prototyping. But, in the interest of cost and ease of assembly, SMD components were found and designed into the resulting board. The circuit consists of 8 sub-circuits (one for each pixel). Each sub-circuit uses two integrated circuits (ICs):

- **MCP4725 DAC** for load application. Uses I²C addresses from 0x60 to 0x67.
- **INA219 ADC** for voltage and amperage monitoring. Uses I²C from 0x40 to 0x47.

In Figure 2, the DAC is used to to apply a certain load voltage to the solar cell. This simulates different load conditions on the solar cell. The DAC can output voltage within the range 0-3V, at a resolution of 0.7mV. This output voltage is the load that is applied to the solar cell. It allows the system to test the solar cell at differing load conditions. The ADC is a high side sensor, which means it should be before the load in the circuit. Its purpose is the measure the voltage and current flowing through the solar cell. It does this by reading the voltage differential across a 10 Ω shunt resistor and using the voltage drop across it to measure current. Using a 10 Ω shunt resistor allows us to measure across a range of 0-32mA with a resolution of 8 μ A. The capacitors (C1, C2, C3) are used to help stabilize power fluctuations in the voltage input.

The inputs and outputs of this circuit are controlled by the ANE over the I²C communication protocol. The use of I²C allows 8 DAC/ADC units to be connected to each board which allows parallel testing.

2.2 Software

This section outlines the data acquisition methods used to assess solar panel stability with our Arduino-based system. The firmware, written in C++, runs on each ANE and communicates with a PC via USB. The PC sends commands specifying the measurement mode and parameters, and each ANE independently returns a continuous data stream that is read by the PC saved as a CSV file for later analysis.

Two modes were implemented to characterize the behavior of solar cells under different load conditions without duplicating the hardware details described above.

2.2.1 JV Scan

In JV scan mode, a controlled voltage sweep—both forward and backward—is applied across the solar cell to generate current-voltage curves. Users can adjust the following parameters:

1. **Scan Range:** The voltage span for the scan (0 V to 3.3 V).
2. **Scan Step Size:** The voltage increment between readings (recommended 0.0001 V to 0.1 V).
3. **Cell Area:** The active area of the solar cell (in mm²), used to calculate current density.
4. **Scan Read Count:** The number of readings at each voltage step (typically 1–10, averaged to reduce noise).
5. **Scan Rate:** The speed of the voltage sweep (0–100 mV/s).

2.2.2 MPPT Algorithm

The MPPT mode employs a perturb-and-observe strategy to dynamically observe the maximum power output. The system starts at a predetermined load voltage, which is applied to all 8 pixels in a device. Starting at a preset load voltage applied to all eight pixels, the system takes periodic measurements of voltage and current, calculates power, and applies a moving average filter to smooth the data. If the smoothed power increases compared to the previous measurement, the voltage is further adjusted in the same direction; if it decreases, the adjustment direction is reversed. If the power decreases, then the direction of voltage adjustment is reversed. The user can configure:

1. **Starting Voltage:** The initial load voltage (0 V to 3.3 V).
2. **Step Size:** The voltage increment for adjustments (recommended 0.0001 V to 0.1 V).
3. **Cell Area:** The active area of the solar cell (in mm²) for live PCE calculations.

4. **Measurements Per Step:** The number of readings per voltage level (typically 5–15, averaged for noise reduction).
5. **Measurement Delay:** The delay between measurements (100–1500 ms) to control data volume.
6. **Measurement Time:** The total duration for the MPPT process (specified in minutes or hours).

2.2.3 Graphical Interface

The Stability Setup GUI is designed to facilitate use of the Stability Setup Hardware. It is built in Python with PySide6, and offers a simple, user-friendly way to manage and control stability tests without having to get into the technical aspects. Here are the functions of the GUI:

1. **Set Up Tests:** Enter details like trial names, email notifications, measurement parameters, etc.
2. **Manage Settings:** Save presets to easily switch between best configurations.
3. **Visualize Data:** Built in live plotting of JV and MPPT curves.
4. **Manage Hardware:** Check how many stability setup systems are connected and configure their IDs.

2.3 Cost and Manufacture

A batch of 5 boards cost approximately \$230. It takes JLCPCB around one to two weeks to produce the boards and deliver to the United States. This price includes shipping and the JLCPCB pick-and-place service which will assemble all surface mount device (SMD) components. ANEs must be purchased separately and assembled with standard 0.1" 2.54mm male breadboard headers. While ANEs are recommended, the system is fully compatible with both official Arduino Nano boards and third party alternatives. During testing, third party Arduino boards performed reliably, making them a viable option for users to further reduce costs without sacrificing performance.

3 Results and Discussion

3.1 Experimental Setup

The Arduino-based system is compared with a commercial MPPT system for validation.

3.2 Data Collection and Analysis

3.2.1 JV Scan

3.2.2 MPPT tracking

3.3 System Accuracy and Reliability

- Comparison of power output data with commercial systems.
- Error analysis and sources of deviation.

3.4 Cost-Effectiveness

- Cost breakdown of the Arduino-based system versus traditional stability measurement equipment.

3.5 Future Improvements

- Integration with more sensors

4 Conclusion

The Arduino-based system provides a cost-effective and reliable alternative for measuring solar panel stability.

5 Supplementary Information

- Github link
- oshwlab link

References

- [1] C. A. R. Perini, T. A. S. Doherty, S. D. Stranks, J.-P. Correa-Baena, and R. L. Z. Hoyer, “Pressing challenges in halide perovskite photovoltaics—from the atomic to module level,” *Joule*, vol. 5, pp. 1024–1030, 2021.